

KCET – 2020

Number of Questions: 60

Total Marks: 120

Mathematics

1. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is

- A. 2^{y-x}
- B. -2^{y-x}
- C. 2^{x-y}
- D. $\frac{2^{y-1}}{2^{x-1}}$

2. If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then $f'(\sqrt{3})$ is

- A. $-\frac{1}{2}$
- B. $\frac{1}{2}$
- C. $\frac{1}{\sqrt{3}}$
- D. $-\frac{1}{\sqrt{3}}$

3. The right hand and left hand limit of the function are respectively

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

- A. 1 and 1
B. 1 and -1
C. -1 and -1
D. -1 and 1
4. If $y = 2x^{n+1} + \frac{3}{x^n}$, then $x^2 \frac{d^2 y}{dx^2}$ is
- A. $6n(n+1)y$
B. $n(n+1)y$
C. $x \frac{dy}{dx} + y$
D. y
5. If the curves $2x=y^2$ and $2xy=K$ intersect perpendicularly, then the value of K^2 is
- A. 4
B. $2\sqrt{2}$
C. 2
D. 8

6. If $(xe)^y = e^y$, then $\frac{dy}{dx}$ is

A. $\frac{\log x}{(1 + \log x)^2}$

B. $\frac{1}{(1 + \log)^2}$

C. $\frac{\log x}{(1 + \log x)}$

D. $\frac{e^x}{x(y-1)}$

7. If the side of a cube is increased by 5%, then the surface area of a cube is increased by

A. 10%

B. 60%

C. 6%

D. 20%

8. The value of $\int \frac{1+x^4}{1+x^6} dx$ is

A. $\tan^{-1} x + \tan^{-1} x^3 + C$

B. $\tan^{-1} x + \frac{1}{3} \tan^{-1} x^3 + C$

C. $\tan^{-1} x - \frac{1}{3} \tan^{-1} x^3 + C$

D. $\tan^{-1} x + \frac{1}{3} \tan^{-1} x^2 + C$

9. The maximum value of $\frac{\log_e x}{x}$, if $x > 0$ is

A. e

B. 1

C. $\frac{1}{e}$

D. $-\frac{1}{e}$

10. The value of $\int e^{\sin x} \sin 2x dx$ is

A. $2e^{\sin x} (\sin x - 1) + C$

B. $2e^{\sin x} (\sin x + 1) + C$

C. $2e^{\sin x} (\cos x + 1) + C$

D. $2e^{\sin x} (\cos x - 1) + C$

11. The value of $\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos^{-1} x dx$ is

A. π

B. $\frac{\pi}{2}$

C. 1

D. $\frac{\pi^2}{2}$

12. If $\int \frac{3x+1}{(x-1)(x-2)(x-3)} dx$ A $\log |x-1|$ B $\log |x-2|$
+ C $\log |x-3|$ + C, then the values of A, B and C
are respectively
- A. 5, -7, -5
B. 2, -7, -5
C. 5, -7, 5
D. 2, -7, 5

13. The value of The value of $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$ is
- A. $\frac{\pi}{2} \log 2$
B. $\frac{\pi}{4} \log 2$
C. $\frac{1}{2}$
D. $\frac{\pi}{8} \log 2$

14. The area of the region bounded by the curve $y^2=8x$ and the line $y=2x$ is

A. $\frac{16}{3} \text{ sq. units}$

B. $\frac{4}{3} \text{ sq. units}$

C. $\frac{3}{4} \text{ sq. units}$

D. $\frac{8}{3} \text{ sq. units}$

15. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^x} dx$ is

A. 2

B. 0

C. 1

D. -2

16. The order of the differential equation obtained by eliminating arbitrary constants in the family of curves $c_1 y = (c_2 + c_3) e^{x+c_4}$

A. 1

B. 2

C. 3

D. 4

17. The general solution of the differential equation

$$x^2 dy - 2xy dx = x^4 \cos x dx \text{ is}$$

- A. $y = x^2 \sin x + cx^2$
- B. $y = x^2 \sin x + c$
- C. $y = \sin x + cx^2$
- D. $y = \cos x + cx^2$

18. The area of the region bounded by the line

$y=2x+1$, x-axis and the ordinates $x=-1$ and $x=1$ is

- A. $\frac{9}{4}$
- B. 2
- C. $\frac{5}{2}$
- D. 5

19. The two vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} + 5\hat{k}$ represent the two sides $\overline{AB}$ and $\overline{AC}$ respectively of a ΔABC . The length of the median through A is

- A. $\frac{\sqrt{14}}{2}$
- B. 14
- C. 7
- D. $\sqrt{14}$

20. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between $\vec{a}$ and $\vec{b}$ then $\sin \frac{\theta}{2}$

A. $|\vec{a} + \vec{b}|$

B. $\frac{|\vec{a} + \vec{b}|}{2}$

C. $\frac{|\vec{a} - \vec{b}|}{2}$

D. $|\vec{a} - \vec{b}|$

21. The curve passing through the point (1, 2) given that the slope of the tangent at any point (x,y) is $\frac{3x}{y}$ represents

A. Circle

B. Parabola

C. Ellipse

D. Hyperbola

22. If $|\vec{a} + \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = 144$ and $|\vec{a}| = 6$ then $|\vec{b}|$ is equal to

A. 6

B. 3

C. 2

D. 4

23. The point $(1, -3, 4)$ lies in the octant
- Second
 - Third
 - Fourth
 - Eighth
24. If the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\lambda\hat{i} - \hat{j} + 2\hat{k}$ are coplanar, then the value of λ is
- 6
 - 5
 - 6
 - 5
25. The distance of the point $(1, 2, -4)$ from the line $\frac{x-3}{2} = \frac{y-3}{2} = \frac{z+5}{6}$ is
- $\frac{293}{7}$
 - $\frac{\sqrt{263}}{7}$
 - $\frac{293}{49}$
 - $\frac{\sqrt{293}}{49}$

26. The sine of the angle between the straight line $\frac{x-2}{3} = \frac{3-y}{-4} = \frac{z-4}{5}$ and the plane $2x-2y+z=5$ is

A. $\frac{3}{\sqrt{30}}$

B. $\frac{3}{50}$

C. $\frac{4}{5\sqrt{2}}$

D. $\frac{\sqrt{2}}{10}$

27. If a line makes an angle of with each of x and and y-axis, then the acute angle made by z-axis is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{6}$

C. $\frac{\pi}{3}$

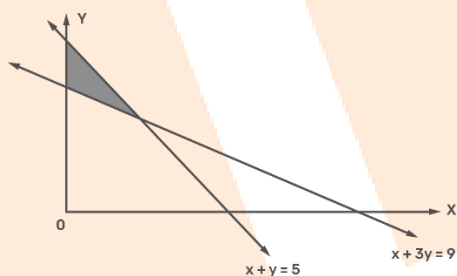
D. $\frac{\pi}{2}$

28. Corner points of the feasible region determined by the system of linear constraints are $(0, 3)$, $(1, 1)$ and $(3, 0)$. Let $z = px + qy$, where $p, q > 0$.

Condition on p and q so that the minimum of z occurs at $(3, 0)$ and $(1, 1)$ is

- A. $p=2q$
- B. $p=q^2$
- C. $p=3q$
- D. $p=q$

29. The feasible region of an LPP is shown in the figure. If $Z = 11x + 7y$, then the maximum value of Z occurs at



- A. $(0,5)$
 - B. $(3,3)$
 - C. $(5,0)$
 - D. $(3,2)$
30. A die is thrown 10 times, the probability that an odd number will come up at least one time is

- A. $\frac{1}{1024}$
- B. $\frac{1023}{1024}$
- C. $\frac{11}{1024}$

D. $\frac{1013}{1024}$

31. If A and B are two events such that

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{2} \text{ and } P(A \cap B) = \frac{1}{6}, \text{ then}$$

$P(A' / B)$ is

A. $\frac{2}{3}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. $\frac{1}{12}$

32. Events E_1 and E_2 from a partition of the sample space S. A is any event such that

$$P(E_1) = P(E_2) = \frac{1}{2}, P(E_2 / A) = \frac{1}{2} \text{ and}$$

$$P(A / E_2) = \frac{2}{3}, \text{ then } P(E_1 / A) \text{ is}$$

A. $\frac{1}{2}$

B. $\frac{2}{3}$

C. 1

D. $\frac{1}{4}$

33. The probability of solving a problem by three persons A, B and C independently is $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{3}$ respectively. Then the probability of the problem is solved by any two of them is

A. $\frac{1}{12}$

B. $\frac{1}{4}$

C. $\frac{1}{24}$

D. $\frac{1}{8}$

34. If $n(A) = 2$ and total number of possible relations from Set A to set B is 1024, then $n(B)$ is

A. 512

B. 20

C. 10

D. 5

35. The value of $\sin^2 51^\circ + \sin^2 39^\circ$ is

A. 1

B. 0

- C. $\sin 12^\circ$
- D. $\cos 12^\circ$

36. If $\tan A + \cot A = 2$, then the value of $\tan^4 A + \cot^4 A =$

- A. 2
- B. 1
- C. 4
- D. 5

37. If $A = \{1, 2, 3, 4, 5, 6\}$, then the number of subsets of A which contain at least two elements is

- A. 64
- B. 63
- C. 57
- D. 58

38. If $z = x + iy$, then the equation $|z+1| = |z-1|$ represents

- A. a circle
- B. a parabola
- C. x-axis
- D. y-axis

39. The value of ${}^{16}C_9 + {}^{16}C_{10} - {}^{16}C_6 - {}^{16}C_7$

- A. 0
- B. 1

C. nC_0

D. ${}^{17}C_3$

40. The number of terms in the expansion of $(x+y+z)^{10}$ is

A. 66

B. 142

C. 11

D. 110

41. If $P(n): 2^n < n!$ Then the smallest positive integer for which $P(n)$ is true if

A. 2

B. 3

C. 4

D. 5

42. The two lines $lx+my=n$ and $l'x+m'y=n'$ are perpendicular if

A. $ll'+mm'=0$

B. $lm'+ml'$

C. $lm+l'm'=0$

D. $lm'+ml'=0$

43. If the parabola $x^2=4ay$ passes through the point (2, 1), then the length of the latus rectum is

A. 1

- B. 4
- C. 2
- D. 8

44. If the sum of n terms of an A.P is given by $S_n = n^2 + n$, then the common difference of the A.P is

- A. 4
- B. 1
- C. 2
- D. 6

45. The negation of the statement "For all real numbers x and y , $x + y = y + x$ " is

- A. For all real numbers x and y , $x + y \neq y + x$
- B. For some real numbers x and y , $x + y = y + x$
- C. For some real number x and y , $x + y \neq y + x$
- D. for some real numbers x and y , $x - y = y - x$

46. The standard deviation of the data 6, 7, 8, 9, 10 is

- A. $\sqrt{2}$
- B. $\sqrt{10}$
- C. 2
- D. 10

47. $\lim_{x \rightarrow 0} \left(\frac{\tan x}{\sqrt{2x + 4} - 2} \right)$ is equal to

- A. 2

- B. 3
- C. 4
- D. 6

48. If a relation R on the set $\{1, 2, 3\}$ be defined by $R = \{(1, 1)\}$, then R is

- A. Reflexive and symmetric
- B. Reflexive and transitive
- C. symmetric and transitive
- D. Only symmetric

49. Let $f: [2, \infty) \rightarrow \mathbb{R}$ be the function defined $f(x) = x^2 - 4x + 5$, then the range of f is

- A. $(-\infty, \infty)$
- B. $[1, \infty)$
- C. $(1, \infty)$
- D. $[5, \infty)$

50. If A, B, C are three mutually exclusive and exhaustive events of an experiment such that $P(A) = 2P(B) = 3P(C)$, then $P(B)$ is equal to

- A. $\frac{1}{11}$
- B. $\frac{2}{11}$
- C. $\frac{3}{11}$

D. $\frac{4}{11}$

51. The domain of the function defined by

$$f(x) = \cos^{-1} \sqrt{x-1}$$

- A. $[1, 2]$
- B. $[0, 2]$
- C. $[-1, 1]$
- D. $[0, 1]$

52. The value of $\cos\left(\sin^{-1} \frac{\pi}{3} + \cos^{-1} \frac{\pi}{3}\right)$ is

- A. 0
- B. 1
- C. -0
- D. Does not exist

53. If $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ then A^4 is equal to

- A. A
- B. 2A
- C. I
- D. 4A

54. If $A = \{a, b, c\}$, then the number of binary operations on A is

- A. A^3
- B. 3^6
- C. 3^3
- D. 3^9

55. If $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then the matrix A is

- A. $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$
- B. $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$
- C. $\begin{pmatrix} -2 & 1 \\ 3 & -2 \end{pmatrix}$
- D. $\begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$

56. If $f(x) = \begin{vmatrix} x^3 - x & a + x & b + x \\ x - a & x^2 - x & c + x \\ x - b & x - c & 0 \end{vmatrix}$ then

- A. $f(1)=0$
- B. $f(2)=0$

- C. $f(0)=0$
D. $f(-1)=0$

57. If A and B are square matrices of same order and B is a skew symmetric matrix, then $A'BA$ is

- A. Symmetric matrix
B. Null matrix
C. Diagonal matrix
D. Skew symmetric matrix

58. If A is a square matrix of order 3 and $|A| = 5$, then $|A \text{ adj.} A|$ is

- A. 5
B. 125
C. 25
D. 625

59. If $f(x) = \begin{cases} \frac{1 - \cos Kx}{x \sin x} & \text{if } x \neq 0 \\ \frac{1}{2} & \text{if } x = 0 \end{cases}$

- A. $\pm \frac{1}{2}$
B. 0
C. ± 2
D. ± 1

60. If $a_1 a_2 a_3 \dots a_9$ are in A.P. then the value of

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} \text{ is}$$

- A. $\frac{9}{2}(a_1 + a_9)$
- B. $(a_1 + a_9)$
- C. $\log_e (\log_e e)$
- D. 1